

Types of Operating Systems

Fundamentals of Operating System Design

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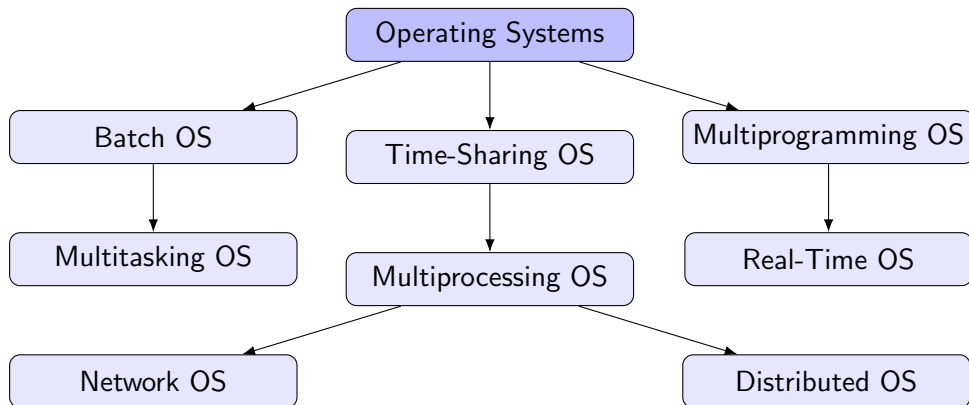
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Learning Outcomes

After completing this lecture, students will be able to:

- Classify different types of operating systems.
- Explain the working principle of each OS type.
- Compare various operating systems.
- Identify suitable applications of each OS.

Classification of Operating Systems



1. Batch Operating System

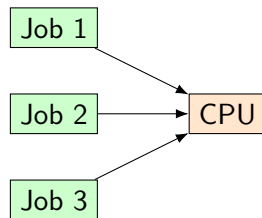
- Jobs are collected into batches.
- No interaction with users during execution.
- Jobs execute sequentially.
- Suitable for repetitive processing.

Advantages

- High throughput
- Simple scheduling

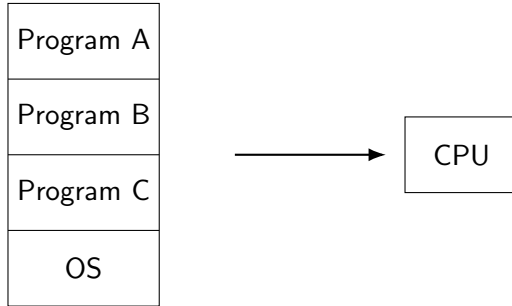
Disadvantages

- Long turnaround time
- No user interaction



2. Multiprogramming Operating System

- Multiple programs reside simultaneously in memory.
- CPU switches when the running process waits for I/O.
- Objective: Maximize CPU utilization.



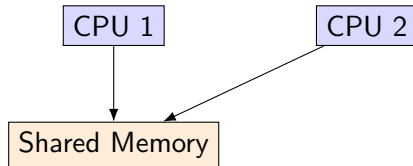
3. Multitasking Operating System

- One user executes multiple applications.
- CPU rapidly switches among tasks.
- Uses Time Quantum.
- Provides fast user response.

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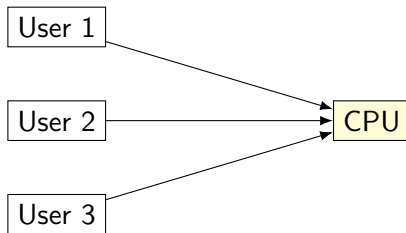
4. Multiprocessing Operating System

- Uses two or more processors.
- True parallel execution.
- High performance and reliability.



5. Time-Sharing Operating System

- Multiple users share the same computer.
- CPU allocates a small time slice to each user.
- Provides interactive computing.



6. Real-Time Operating System (RTOS)

Definition

An RTOS guarantees that tasks are completed within specified timing constraints.

- Deterministic response
- Very low latency
- High reliability
- Used in embedded systems

Applications

- Medical equipment
- Robotics
- Industrial automation
- Aircraft control systems

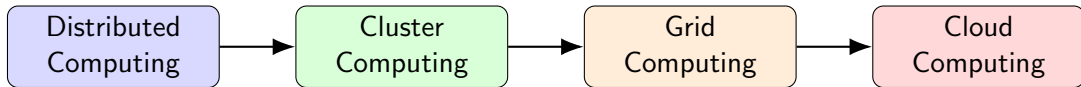
7. Network Operating System

- Manages computers connected through a network.
- Supports file sharing.
- Supports printer sharing.
- Provides centralized administration.

8. Distributed Operating System

- Multiple computers work as one system.
- Transparent resource sharing.
- Load balancing.
- Fault tolerance.

Evolution of Distributed Computing



- All use multiple computers.
- They differ in ownership, location, management, and resource sharing.
- Cloud computing evolved from distributed computing concepts.

Cluster Computing

Definition

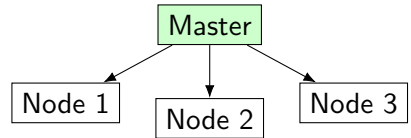
A cluster is a collection of interconnected computers that work together as a single system.

Characteristics

- Same organization
- High-speed LAN
- Centralized management
- High performance
- High availability

Applications

- Scientific Computing
- AI Training
- Database Servers

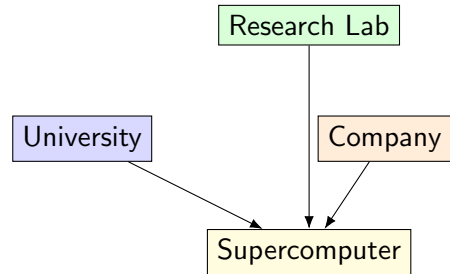


Definition

Grid computing combines geographically distributed computers from multiple organizations to solve large computational problems.

Characteristics

- Different organizations
- Geographically distributed
- Heterogeneous hardware
- Resource sharing
- Large-scale computation



Applications

- Weather forecasting
- Protein analysis
- Particle physics

Cloud Computing

Definition

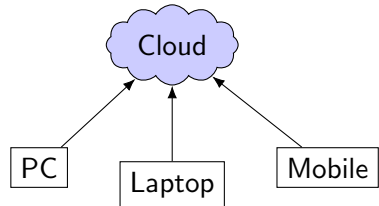
Cloud computing provides computing resources such as servers, storage, databases, and software over the Internet on demand.

Characteristics

- Pay-as-you-go
- Elastic scalability
- Virtualization
- On-demand services
- Accessible anywhere

Service Models

- IaaS
- PaaS
- SaaS



Comparison of Computing Paradigms

Feature	Cluster	Grid	Cloud	Distributed
Ownership	Single	Multiple	Cloud Provider	Multiple
Location	Same Site	Different Sites	Internet	Network
Hardware	Usually Similar	Heterogeneous	Virtualized	May Differ
Management	Centralized	Distributed	Provider Managed	Distributed
Primary Goal	High Performance	Resource Sharing	On-demand Services	Communication
Examples	HPC Cluster	Scientific Grid	AWS, Azure, GCP	Distributed OS

Comparison of Operating Systems

Type	Main Goal	Example
Batch	Automatic job execution	IBM Mainframe
Multiprogramming	CPU utilization	Linux
Multitasking	User responsiveness	Windows
Multiprocessing	Parallel execution	Linux Server
Time-Sharing	Multi-user interaction	UNIX
Real-Time	Predictable response	VxWorks
Network	Resource sharing	Windows Server
Distributed	Distributed computing	Amoeba

Key Takeaways

- Batch OS executes jobs in batches.
- Multiprogramming maximizes CPU utilization.
- Multitasking improves user experience.
- Multiprocessing uses multiple CPUs.
- Time-sharing supports multiple users.
- RTOS guarantees timely responses.
- Network and Distributed OS enable resource sharing across computers.